Geometric Limits on Semantic Expressibility: Quantum-Inspired Contexts and Polysemy in Large Language Models

Karl Svozil[1,*]

[1] *Institute for Theoretical Physics, TU Wien, Wiedner Hauptstraße 8–10/136, 1040 Vienna, Austria*

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High-dimensional embedding spaces can host many semantic directions with small mutual overlap. But small overlaps are not zero: when many directions are jointly active, their residual interference accumulates and limits what a finite readout channel can recover. We formulate this as a distinction between *kinetic capacity*—what the geometry can host—and *epistemic accessibility*—what readout can recover. The two sides are summarized by $N \lesssim \exp(c\, d_{\mathrm{eff}}\, \varepsilon^2)$ for coexistence and $\sigma_{\mathrm{int}} \sim \sqrt{k/d_{\mathrm{eff}}}$ for simultaneous readout. Thus dimension acts not merely as storage capacity but as semantic concurrency bandwidth. On this geometric foundation we propose a separate hypothesis: some polysemous tokens may be organized around stable token-associated reference directions, with sense information carried by low-dimensional subspaces in their orthogonal complement. The capacity/accessibility distinction is the main claim; the reference-subspace hypothesis is a stronger, separately falsifiable empirical proposal.

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I. INTRODUCTION

This chapter examines contextual representation in large language models (LLMs) as part of the volume's broader inquiry into "quantum thinking." Only a minimal orientation to LLM operation is needed here. Input text is first divided into tokens, and each token is mapped to an initial vector. Transformer blocks then use self-attention and feed-forward transformations to turn those initially context-free vectors into sequence-dependent hidden states. An output layer converts the final state into token scores, and a softmax distribution supports probabilistic next-token selection. This compact chain—tokens, embeddings, contextual transformation, and finite readout—supplies the computational background for the argument below; the details of pre-training, fine-tuning, reinforcement learning, and backpropagation are not required.

Current Transformer-based LLMs are classical computational systems. They run on classical hardware and use real-valued matrix operations, normalization, nonlinearities, copying, and generally irreversible updates. Their reliance on vectors, inner products, high dimension, or probabilistic output does not by itself make them quantum or even quantum-like. The more limited question pursued here is where ordinary high-dimensional geometry already explains the coexistence and contextual selection of semantic features, and where a stronger comparison with quantum-like formalisms might begin. Such a comparison would have to do more than rename familiar neural operations: it would need to identify a role for contextual bases or subspaces, incompatible or noncommuting alternatives, interference, or measurement-like readout that yields explanatory or predictive value beyond a classical dynamic-embedding account.

Semantic concurrency makes this boundary concrete. A model must preserve many latent possibilities while making only a small, context-selected subset operationally accessible. Polysemy is the simplest illustration: the same token can participate in the meanings of "river bank" and "financial bank," yet the surrounding sequence drives it toward different contextual representations. This relation among possibility, context, and readout connects directly to the volume's recurring themes without implying that the underlying machine is physically quantum. It also motivates a separation between what a representation can geometrically accommodate and what a finite channel can reliably recover.

Within Part III, the chapter therefore occupies a deliberately intermediate position. Chapter 3 supplies the shared vocabulary of state spaces, contexts, Hilbert-space geometry, and quantum-like representation. Chapter 7 develops a contrasting route in which classical oscillatory and wave dynamics can support interference-like computation. The present chapter instead asks how far high-dimensional representation, contextual selection, and readout can take a classical artificial language system. Chapter 9 places these results beside neural systems, embodiment, meaning, and the limits of the quantum analogy. Chapter 13 and Bridge C provide the corresponding standards for model comparison and evidence: an analogy should be sharpened into explicit structure and then into a discriminative test before it is treated as an explanation.

The claims below consequently occupy different rungs of that evidential ladder. Concentration of measure and exponential packing are standard results of high-

* karl.svozil@tuwien.ac.at

dimensional geometry. Their use to describe semantic capacity, together with the accumulated-interference estimate, is a formal model of classical representation subject to stated assumptions about effective dimension and coactivation. The reference-subspace organization is a stronger empirical hypothesis about LLM internals and is tested against classical controls. The comparison with quantum contextuality remains a limited structural correspondence; no physical quantumness, Born-rule dynamics, or semantic Kochen–Specker theorem is inferred.

A finite-dimensional activation space faces a simple tension. High dimension lets many semantic features coexist as nearly orthogonal directions. Finite readout prevents too many of them from being simultaneously recovered from one channel. This chapter makes that tension quantitative.

The two sides obey different scalings. On the capacity side, concentration of measure permits exponentially many candidate semantic directions with pairwise overlap at most ε,

$$N \lesssim \exp(c\, d_{\text{eff}}\, \varepsilon^2). \tag{1}$$

On the readout side, the same small residual overlaps add in quadrature over an active set of size k, giving an interference scale

$$\sigma_{\text{int}} \sim \sqrt{k/d_{\text{eff}}}. \tag{2}$$

The first relation is an existential packing estimate. The second is a typical-case readout-noise estimate. Together they say that dimension is a *semantic concurrency bandwidth*: it does not mainly limit how many meanings can be geometrically stored, but how many independently recoverable semantic components can be jointly active.

Polysemy is the cleanest example. A single lexical item, such as "bank", can access several mutually exclusive meanings, only one of which is selected in a given context. The same tension appears more generally in feature superposition, role binding, and compositional semantics [1]: many latent semantic degrees of freedom must coexist, while only a context-selected subset is read out. We use polysemy as a controlled entry point because it makes contextual selection explicit. In lexical semantics, polysemy is distinguished from homonymy by the relatedness of senses [2]. Modern Transformer-based LLMs implement contextual selection through dynamic contextual embeddings: a token representation is repeatedly transformed by self-attention and MLP layers as it absorbs information from its sentential environment [3].

The chapter makes two claims. The first is the geometric capacity claim: concentration permits large semantic capacity, while accumulated interference limits simultaneous accessibility. This claim is architecture-independent. The second is a stronger organizational hypothesis: for some canonically polysemous tokens, readout may be structured relative to a stable token-associated *reference direction*, with sense information carried by low-dimensional subspaces in its orthogonal complement. The reference-subspace hypothesis is not implied by concentration geometry. It is an empirical proposal, and Sec. V D states how it can fail.

A serious caveat is effective dimension. Trained representations are often anisotropic: they occupy a narrow cone rather than the full ambient space. The dimension controlling concentration is therefore not necessarily the raw embedding dimension d, but an effective dimension d_{eff}. If d_{eff} is small, the capacity claim weakens accordingly. For example, at $d_{\text{eff}} \sim 50$ and $\varepsilon = 0.1$, the exponent $c d_{\text{eff}} \varepsilon^2$ is only a small constant. The framework therefore applies to the approximately isotropic geometry actually used by the representation, possibly after whitening or restriction to a participation subspace. Determining which layers, models, and preprocessing regimes admit a meaningful d_{eff} is an empirical priority.

The reference-subspace hypothesis has a narrower domain. It is meant for canonically polysemous tokens with established sense inventories, such as WordNet-, OntoNotes-, SemCor-, or WiC-style sense distinctions [4] with enough contextual instances for testing. Failure on a representative sample, against the controls in Sec. V D, would count against a privileged token reference direction even if the broader capacity/accessibility picture survives.

The proposal is adjacent to the superposition hypothesis in mechanistic interpretability [5], according to which neural networks represent more features than they have neurons by packing features into nearly orthogonal directions. The present chapter separates the geometric background from a possible semantic organization within it. The general superposition picture concerns an overcomplete dictionary of feature directions and sparse active sets. The reference-subspace picture adds the more specific possibility that some contextual readout is organized around stable token-associated directions.

Section II develops the capacity framework: quasi-orthogonality, Lévy's lemma, exponential packing, and the kinetic/epistemic distinction. Section III introduces the reference-subspace hypothesis and its relation to dynamic contextual embeddings. Section IV connects the picture to attention as soft routing rather than literal basis reconstruction. Section V discusses superposition, sparse autoencoders, limitations, empirical tests, and the limited quantum analogy.

Notation. Throughout, $\mathbb{R}^d$ is Euclidean d-space with inner product $u^\mathsf{T} v$ and norm $\|v\| = \sqrt{v^\mathsf{T} v}$; $S^{d-1} = \{v \in \mathbb{R}^d : \|v\| = 1\}$ is the unit sphere; I is the identity matrix; $\|\cdot\|_{\text{op}}$ is the operator norm; $A \succeq 0$ means positive semidefinite; and $\mathbb{P}$ and $\mathbb{E}$ denote probability and expectation under the uniform measure on the relevant sphere unless stated otherwise. Universal positive constants are denoted c, c', possibly changing from line to line. The symbol d_{eff} denotes the effective dimension controlling concentration in the representation being probed.

II. CONCENTRATION GEOMETRY AND CAPACITY

This section is architecture-independent. It gives the geometric capacity constraint on which the later reference-subspace hypothesis builds.

A. Quasi-orthogonality

Let $w_1, \ldots, w_N \in \mathbb{R}^d$ be unit feature directions. Strict orthogonality, $w_i^\mathsf{T} w_j = 0$ for $i \neq j$, allows at most $N = d$ such vectors. Semantic encoding does not require exact orthogonality. It requires approximate distinguishability. We therefore call a family ε-quasi-orthogonal if

$$|w_i^\mathsf{T} w_j| \leq \varepsilon, \qquad i \neq j. \tag{3}$$

This is the relevant notion for high-dimensional feature storage: small overlap is tolerable, provided readout can absorb the resulting interference.

B. Lévy's lemma and linear overlaps

High-dimensional spheres make quasi-orthogonality generic.

The mechanism is already visible before invoking the formal lemma. Let

$$x = (x_1, \ldots, x_d) \in \mathbb{R}^d$$

be a unit vector chosen without a preferred direction. Then

$$x_1^2 + \cdots + x_d^2 = 1.$$

Rotational invariance suggests that the norm is spread over the available coordinates, so a typical coordinate weight is of order $1/d$. Now fix a unit direction u. By rotational invariance one may choose coordinates so that $u = e_1$. The squared overlap with the random direction is then

$$|u^\mathsf{T} x|^2 = x_1^2.$$

Thus the overlap with any fixed direction is typically of order $d^{-1/2}$ in amplitude, or $1/d$ in squared amplitude. A macroscopic overlap $|u^\mathsf{T} x| > \varepsilon$ therefore requires one coordinate to carry an anomalously large fraction of the total norm. In high dimension such an event is exponentially unlikely.

The same intuition appears particularly transparently in the complex Hilbert-space analogue. If $\psi \in \mathbb{C}^D$ is Haar-random and ϕ is fixed, unitary invariance permits the choice $\phi = e_1$, so

$$|\langle \phi, \psi \rangle|^2 = |\psi_1|^2.$$

For a Haar-random complex unit vector the exact tail is

$$\mathbb{P}\{|\langle \phi, \psi \rangle|^2 \geq \eta\} = (1-\eta)^{D-1} \leq \exp[-(D-1)\eta].$$

For real embedding vectors the analogous spherical concentration estimate has the operative scaling

$$\mathbb{P}\{|u^\mathsf{T} x| > \varepsilon\} \lesssim \exp(-cd\varepsilon^2)$$

for a universal constant $c > 0$. Consequently, if N random directions are sampled, a union-bound estimate gives a probability at most of order $N^2 \exp(-cd\varepsilon^2)$ that some pair has overlap larger than ε. Hence N may grow exponentially in $d\varepsilon^2$ while all pairwise overlaps remain small with high probability. This is the heuristic core of the exponential quasi-orthogonal capacity derived below.

Lévy's lemma makes this intuition precise. If $f : S^{d-1} \to \mathbb{R}$ is L-Lipschitz, then

$$\mathbb{P}\big(|f(v) - \mathbb{E}f| \geq \delta\big) \leq 2\exp\left(-\frac{c\,d\,\delta^2}{L^2}\right), \tag{4}$$

for a universal $c > 0$ [6, 7]. Apply this to the linear functional $f_u(v) = u^\mathsf{T} v$. It is 1-Lipschitz and has mean zero by symmetry. Hence

$$\mathbb{P}\big(|u^\mathsf{T} v| \geq \varepsilon\big) \leq 2\exp(-c\,d\,\varepsilon^2). \tag{5}$$

For normalized surface measure one obtains the explicit form $2\exp(-(d-1)\varepsilon^2/2)$ [6]. Thus two random unit vectors are nearly orthogonal with high probability, and typical overlap amplitudes scale as $d^{-1/2}$.

In trained models, d should be replaced by d_{eff}, the dimension actually controlling concentration. Anisotropy, low intrinsic dimension, or a narrow activation cone reduce d_{eff}. Whitening or projection to a participation subspace may increase the effective dimension relevant for the analysis.

Relation to Johnson–Lindenstrauss. The Johnson–Lindenstrauss lemma preserves pairwise distances among n points after projection to dimension $k = O(\varepsilon^{-2}\log n)$ [8]. It is driven by the same concentration phenomenon as Eq. (5); indeed JL can be derived from sphere concentration applied to projection norms [9]. Here Lévy's lemma is the more direct tool: the issue is not projecting a finite dataset, but the typical overlap between semantic directions in the representation space.

C. Exponential packing

Sample N unit vectors independently and uniformly from S^{d-1}. By Eq. (5) and a union bound over pairs,

$$\mathbb{P}\big(\exists\, i \neq j :\ |w_i^\mathsf{T} w_j| \geq \varepsilon\big) < N^2 \exp(-c\,d_{\text{eff}}\,\varepsilon^2). \tag{6}$$

This probability is below one, and hence an ε-quasi-orthogonal family exists, provided

$$N < \exp\left(\frac{c}{2}\,d_{\text{eff}}\,\varepsilon^2\right). \tag{7}$$

This is the kinetic capacity estimate: in high effective dimension, many candidate semantic directions can coexist with small pairwise overlap.

The estimate controls pairwise overlap only. It does not guarantee that a large collection of feature directions is well-conditioned. By Gershgorin, $|w_i^\top w_j| \leq \varepsilon$ gives only $\|G - I\|_{\mathrm{op}} \leq (n-1)\varepsilon$ for an n-element Gram matrix, which is useless when n is large. Joint readout therefore needs a separate conditioning assumption, introduced in Sec. III B.

D. Kinetic capacity versus epistemic accessibility

The packing bound says what can coexist. It does not say what can be read. We call the coexistence supply *kinetic semantic capacity*. We call the recoverable subset *epistemic semantic accessibility*. The distinction is the central point of the chapter.

Let N candidate features be represented by unit directions $w_1, \ldots, w_N \in \mathbb{R}^{d_{\mathrm{eff}}}$. In a given context, suppose an active set S of size k contributes amplitudes x_j, producing

$$h = \sum_{j \in S} x_j w_j. \tag{8}$$

A linear readout of feature i gives

$$w_i^\top h = x_i + \sum_{\substack{j \in S \\ j \neq i}} x_j \, w_i^\top w_j. \tag{9}$$

The first term is signal; the second is interference. For random incoherent directions,

$$\mathbb{E}[w_i^\top w_j] = 0, \qquad \mathrm{Var}(w_i^\top w_j) \simeq \frac{1}{d_{\mathrm{eff}}}. \tag{10}$$

Thus, conditional on the active amplitudes,

$$\mathrm{Var}\left(\sum_{\substack{j \in S \\ j \neq i}} x_j \, w_i^\top w_j \right) \simeq \frac{1}{d_{\mathrm{eff}}} \sum_{\substack{j \in S \\ j \neq i}} x_j^2. \tag{11}$$

For amplitudes of order unity,

$$\sigma_{\mathrm{int}} \sim \sqrt{\frac{k}{d_{\mathrm{eff}}}}. \tag{12}$$

This is the readout bottleneck. Concentration lets many directions coexist because their overlaps are small. But readout sees the *sum* of many small overlaps. Reliable recovery requires

$$\sqrt{\frac{k}{d_{\mathrm{eff}}}} \ll \Delta, \tag{13}$$

where Δ is the relevant decision margin. Equivalently,

$$\text{kinetic: } N \lesssim \exp(c \, d_{\mathrm{eff}} \, \varepsilon^2), \qquad \text{epistemic: } k \lesssim d_{\mathrm{eff}} \, \Delta^2. \tag{14}$$

The first is storage-like coexistence. The second is simultaneous accessibility. A model may host many more features than it can jointly read.

Figure 1 gives a minimal synthetic illustration of this distinction. The simulation is not intended as a model of a trained language model. It is only a null model for the concentration geometry: random unit directions in an effective dimension d_{eff} can coexist in large quasi-orthogonal families, while simultaneous linear readout accumulates the residual overlaps of the active directions. Deviations from this null model in trained representations would themselves be informative, reflecting anisotropy, learned feature alignment, nonlinear readout, or coactivation-dependent orthogonalization.

This also clarifies the relation to compressed sensing. Stable recovery of a k-sparse vector in an N-dimensional feature dictionary from m measurements typically requires $m \gtrsim Ck \log(N/k)$ under suitable incoherence or restricted-isometry assumptions [10–12]. In an idealized isotropic residual stream, one may identify m with d_{eff}; in trained models the effective representation dimension and effective readout dimension may differ. The compressed-sensing analogy is therefore diagnostic, not literal.

III. TOKEN REFERENCE DIRECTIONS AND SENSE SUBSPACES

The previous section established an architecture-independent capacity constraint. We now ask a more speculative question: does a trained model organize the contextual states of some polysemous tokens relative to a stable token-associated reference subspace? This hypothesis is not implied by concentration geometry, and it is separately falsifiable.

The notation distinguishes mathematical types and roles. For a token t and a sense context C:

- $r_t \in \mathbb{R}^d$ is a unit candidate token reference vector;
- $\mathcal{R}_t \subset \mathbb{R}^d$ is the token reference subspace;
- $S_{t,C} \subset \mathcal{R}_t^\perp$ is a low-dimensional sense subspace;
- $u_{t,C,j} \in S_{t,C}$ are vectors spanning that sense subspace;
- $h_\ell(t \mid x) \in \mathbb{R}^d$ is an observed contextual hidden state.

Thus r_t, $u_{t,C,j}$, and $h_\ell(t \mid x)$ are vectors, whereas $\mathcal{R}_t$ and $S_{t,C}$ are subspaces. The w_i used in Sec. II have a separate role: they are generic feature directions in the capacity calculation and are not reused as sense labels.

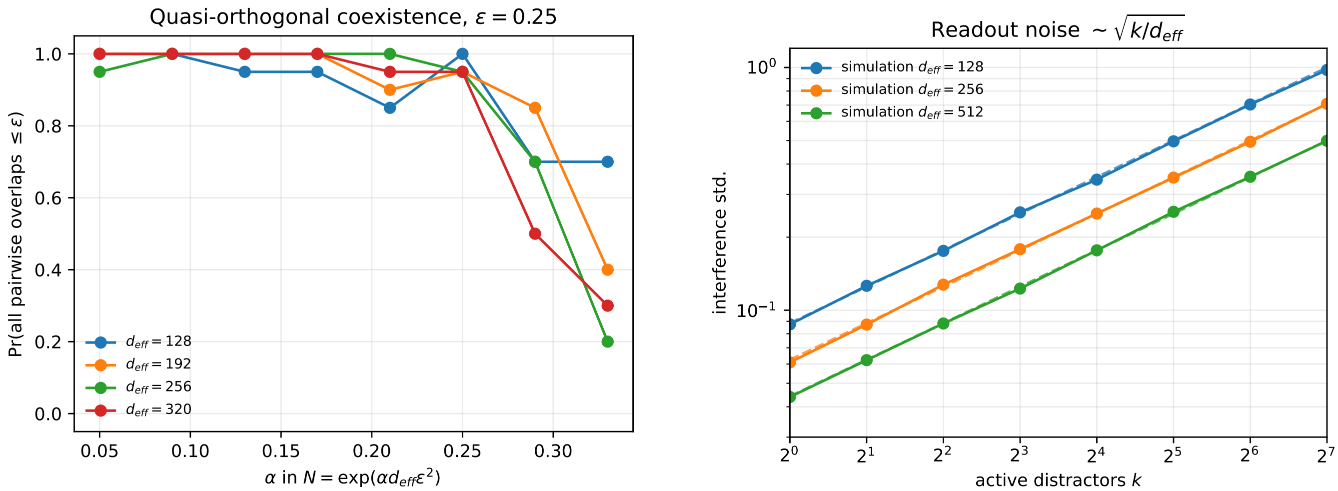

FIG. 1. Synthetic geometry checks for the capacity/accessibility distinction. Left: random unit directions in dimension d_{eff} are sampled with $N = \exp(\alpha\, d_{\text{eff}}\, \varepsilon^2)$ and tested for ε-quasi-orthogonality. For sufficiently small α, large families coexist with high probability, illustrating the exponential packing side of Eq. (14). Right: for a fixed target direction, k random distractor directions are simultaneously active with random signs. The standard deviation of the readout interference follows the predicted scale $\sigma_{\text{int}} \sim \sqrt{k/d_{\text{eff}}}$, illustrating the accessibility side of Eq. (14). Both panels use random isotropic vectors and should be read as a geometric null model rather than as measurements on a trained language model.

A. The reference-subspace hypothesis

In the simplest version, let r_t be a candidate unit direction and set

$$\mathcal{R}_t = \operatorname{span}\{r_t\}. \tag{15}$$

Empirically, r_t could be the normalized static input embedding, a normalized mean contextual embedding, or a learned token-specific probe direction. It is simply a hypothesized context-stable reference for token identity. It is not a mechanical or dynamical object. Any state can be decomposed relative to any direction, so the decomposition alone has no empirical content. The claim is that a token-associated $\mathcal{R}_t$ exposes sense structure better than matched control directions or subspaces.

For each context or sense label C, define a low-dimensional subspace

$$S_{t,C} = \operatorname{span}\{u_{t,C,1}, \ldots, u_{t,C,q_C}\} \subset \mathcal{R}_t^{\perp}, \qquad q_C \ll d_{\text{eff}}. \tag{16}$$

For example, finance, river-edge, and aircraft-tilt uses of "bank" may correspond to different subspaces $S_{\text{bank},C}$ inside the common orthogonal carrier $\mathcal{R}_{\text{bank}}^{\perp}$. A word sense is not a single vector: topic, syntax, selectional preferences, and world knowledge may require several spanning directions $u_{t,C,j}$.

The reference object may itself require more than one direction. In that case define

$$\mathcal{R}_t = \operatorname{span}\{r_{t,1}, \ldots, r_{t,p}\}, \qquad p \geq 1, \tag{17}$$

and require

$$S_{t,C} \subset \mathcal{R}_t^{\perp}. \tag{18}$$

Equation (15) is the special case $p = 1$. The analogy with interlinked quantum contexts is limited to a pattern of sharing. In quantum logic, different contexts can share one or more observables or projections; finite-dimensional examples with two shared observables are known [13]. In the semantic model, $\mathcal{R}_t$ is instead a common token-associated reference subspace, and the sense-specific subspaces lie in its orthogonal complement. No noncommutativity or quantum probability is implied by Eqs. (17)–(18).

The low-rank assumption is empirical. It predicts that dominant within-sense variation is low-dimensional after projection onto $\mathcal{R}_t^{\perp}$. This is compatible with evidence that contextualized embeddings can separate word senses, but it is stronger: it asks whether such separation is privileged relative to a token-associated reference subspace [14, 15].

B. Projection and gated subspace readout

Let $h(x) \in \mathbb{R}^d$ be a contextual state and let P_t be the orthogonal projector onto $\mathcal{R}_t$. The two components are

$$\begin{aligned} h(x) &= h_{\text{ref}}(x) + h_{\perp}(x), \\ h_{\text{ref}}(x) &= P_t h(x), \\ h_{\perp}(x) &= (I - P_t) h(x) \in \mathcal{R}_t^{\perp}. \end{aligned} \tag{19}$$

For the one-dimensional case, $P_t = r_t r_t^{\mathsf{T}}$ and the scalar $a_t(x) = r_t^{\mathsf{T}} h(x)$ measures the component along the reference direction. The hypothesis is that most sense-disambiguating variation is present in $h_{\perp}(x)$, not that the projection formula itself is special.

Readout inside $S_{t,C}$ also requires the spanning vectors to be well-conditioned. Let

$$U_{t,C} = [\, u_{t,C,1} \ \cdots \ u_{t,C,q_C} \,], \qquad G_{t,C} = U_{t,C}^{\mathsf{T}} U_{t,C}. \quad (20)$$

The columns of $U_{t,C}$ span $S_{t,C}$; $U_{t,C}$ is a matrix, not an additional semantic vector. We call it ρ-well-conditioned if

$$\|G_{t,C} - I\|_{\mathrm{op}} \le \rho < 1. \quad (21)$$

Then

$$\widetilde{U}_{t,C} = U_{t,C} G_{t,C}^{-1} \quad (22)$$

is a dual spanning matrix and satisfies $\|\widetilde{U}_{t,C}\|_{\mathrm{op}} \le (1-\rho)^{-1/2}$. Near-degenerate spanning sets amplify readout noise, so the model requires ρ comfortably below one. The corresponding coordinates are

$$\alpha_{t,C}(x) = \widetilde{U}_{t,C}^{\mathsf{T}} h_\perp(x) \in \mathbb{R}^{q_C}. \quad (23)$$

A simple sense score is

$$s_{t,C}(x) = g_t(a_t(x))\, f_{t,C}(h_\perp(x)), \quad (24)$$

where $g_t \ge 0$ gates engagement with the token reference direction and $f_{t,C} \ge 0$ measures fit to the sense subspace. One possible choice is $f_{t,C} = \|\alpha_{t,C}(x)\|^2$; a learned positive quadratic form is another. The selected sense label is

$$C^*(x) = \arg\max_C s_{t,C}(x). \quad (25)$$

The empirical content is the proposed separation of roles: the reference component tracks token identity or engagement, while the orthogonal component contains sense-disambiguating information. If no token-associated $\mathcal{R}_t$ exposes sense structure better than matched random, frequency-controlled, or layer-matched subspaces, the hypothesis fails.

Figure 2 illustrates these objects without introducing extra vector symbols.

C. Packing sense subspaces in a shared orthogonal carrier

The vector packing bound extends to low-dimensional subspaces. Fix a unit reference vector $r_0 \in S^{d-1}$ and let $m = d-1$. For $1 \le q \ll m$ and any fixed threshold $\mu > 2\sqrt{q/m}$, there exist q-dimensional subspaces $S_1, \ldots, S_M \subset r_0^\perp$ such that

$$\|U_a^{\mathsf{T}} U_b\|_{\mathrm{op}} \le \mu, \qquad a \ne b, \quad (26)$$

where $U_a, U_b \in \mathbb{R}^{m\times q}$ are orthonormal basis matrices, and

$$M \ge \exp(c_{\mu,q} m) \quad (27)$$

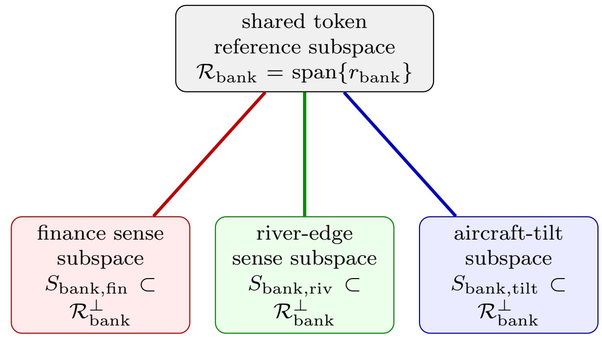

FIG. 2. Organization proposed by the reference-subspace hypothesis for the polysemous token "bank." This is a relationship diagram, not a coordinate plot. The central box is the one-dimensional reference subspace $\mathcal{R}_{\text{bank}} = \text{span}\{r_{\text{bank}}\}$; the three outer boxes are entire low-dimensional sense subspaces, not individual vectors. All three lie in the common carrier $\mathcal{R}_{\text{bank}}^\perp$. In the multi-direction version, $\mathcal{R}_{\text{bank}}$ has dimension $p > 1$, but the carrier remains its orthogonal complement. The undirected lines indicate this common-reference relationship; they do not denote maps or causal flow.

for some positive $c_{\mu,q}$.

This is a standard Grassmannian packing consequence of concentration on Stiefel and Grassmann manifolds [16, 17]. For two independent Haar-random q-planes, the singular values of $U^{\mathsf{T}}V$ are the cosines of their principal angles. In the regime $q \ll m$, the largest principal correlation is of order $\sqrt{q/m}$ [18–20]. A union bound over sampled subspaces then gives an exponential packing.

For trained representations this statement must be read inside an approximately isotropic effective carrier. If $\dim(\mathcal{R}_t) = p$, its effective dimension is $m_{\text{eff}} = d_{\text{eff}} - p$, not necessarily the raw ambient dimension minus p. Without such an effective carrier, the Haar-random subspace model is not a good description of activation geometry.

D. Relation to dynamic contextual embeddings

Transformers already handle polysemy dynamically. The hidden state $h_\ell(t \mid x)$ of token t at layer ℓ depends on context x. The reference-subspace hypothesis is not an alternative architecture; it is a proposed organization of these observed trajectories.

Given a candidate reference subspace $\mathcal{R}_t$ and its projector P_t, write

$$\begin{aligned} h_\ell(t \mid x) &= P_t h_\ell(t \mid x) + z_\ell(t,x), \\ z_\ell(t,x) &= (I - P_t) h_\ell(t \mid x) \in \mathcal{R}_t^\perp. \end{aligned} \quad (28)$$

The decomposition is automatic; the hypothesis is not. It predicts that $P_t h_\ell(t \mid x)$ tracks token identity or engagement, while $z_\ell(t,x)$ carries most sense-disambiguating information. Sense labels should be more separable in $\mathcal{R}_t^\perp$ than in the orthogonal complements of matched control subspaces. For each sense C, the projected states $z_\ell(t,x)$

should also concentrate near the low-dimensional subspace $S_{t,C} \subset \mathcal{R}_t^{\perp}$.

Thus the critical question is not whether contextual embeddings move; they do. It is whether their sense-relevant movement has a privileged organization relative to a token-associated reference subspace.

IV. ATTENTION AND LATENT CONTEXT SELECTION

A standard attention head computes

$$h' = \sum_j \alpha_j \nu_j, \qquad \alpha_j = \frac{\exp(q^\intercal k_j/\sqrt{d})}{\sum_{j'} \exp(q^\intercal k_{j'}/\sqrt{d})}. \tag{29}$$

Keys and values are produced by different learned maps, so value vectors are not generally coefficients of a dual basis for the key vectors. Attention is therefore not literal basis reconstruction.

The useful analogy is softer. Attention routes information: it amplifies some contextual directions and suppresses others. In the terminology of Sec. III, this resembles context-dependent subspace selection. A literal basis-reconstruction interpretation would require additional constraints, such as approximately Parseval keys and values implementing the corresponding dual reconstruction. We do not assume this.

A compact latent-variable version is

$$p(y \mid x, t) = \sum_C p(y \mid x, t, C)\, p(C \mid x, t), \tag{30}$$

where t is the token being disambiguated, y is a possible output token, and C indexes candidate sense contexts. One may model

$$p(C \mid x, t) = \frac{\exp(s_{t,C}(x)/T)}{\sum_{C'} \exp(s_{t,C'}(x)/T)}, \tag{31}$$

with $s_{t,C}$ the gated score of Eq. (24). This is not a claim about the literal implementation of decoding in current LLMs: standard temperature acts on token logits, not on explicit latent sense labels. The point is only that ambiguity can be represented as a distribution over possible sense contexts rather than as a single fixed interpretation.

V. DISCUSSION

The main proposal is the capacity/accessibility distinction. The reference-subspace hypothesis is one possible organization within that constraint, not a consequence of it.

A. Relation to superposition and sparse autoencoders

The superposition hypothesis holds that neural networks represent more features than they have neurons by packing features into nearly orthogonal directions [5, 21]. The concentration estimate (5) gives the kinetic side of that picture: many directions can coexist. Equation (12) gives the epistemic side: many jointly active features create interference of scale $\sqrt{k/d_{\text{eff}}}$. Sparsity is therefore not an implementation detail; it is necessary for readout.

Sparse autoencoders (SAEs) give a concrete empirical interface. They approximate layer activations as sparse expansions over learned decoder directions, making the w_i of Eq. (8) measurable objects rather than hypothetical features. Recent work has scaled SAEs, studied SAE feature geometry, and automated interpretation of large numbers of latents [22–25]. In the present framework, SAE decoder directions can test the capacity/accessibility prediction, while sense-conditioned SAE activation clusters can test the stronger reference-subspace hypothesis.

B. A coactivation-weighted capacity limit

The active-set size k is a crude summary. A more refined description uses the coactivation graph

$$A_{ij} = \mathbb{P}(i, j \text{ active}). \tag{32}$$

If active amplitudes have, conditional on coactivation, zero mean and pairwise uncorrelated signs, then cross terms cancel in expectation and feature j contributes to the readout variance of feature i in proportion to $\mathbb{E}[x_j^2 \mid i, j \text{ active}](w_i^\intercal w_j)^2$. Absorbing amplitude variance and feature importance into coefficients b_i, b_j gives

$$E_{\text{int}} = \sum_{i \neq j} A_{ij} b_i b_j (w_i^\intercal w_j)^2. \tag{33}$$

Frequently coactive features are therefore pressured toward orthogonality. Rarely coactive or mutually exclusive features are much less constrained and may share directions or even become approximately antipodal if nonlinear readout makes this useful. The relevant object is not just the number of features, but the weighted coactivation graph.

C. Limitations

The framework can fail in several ways.

First, anisotropy may make d_{eff} much smaller than d. Contextual embeddings are known to be anisotropic, and intrinsic dimension can vary strongly by layer and training regime [26–29]. Whitening and related postprocessing can partially restore isotropic geometry [27, 28, 30]. At the same time, anisotropy should not be treated as an automatic or universal property of all Transformer representations [31]. The bounds should always be read in terms of the effective dimension actually achieved.

TABLE I. Dynamic contextual embeddings versus the reference-subspace hypothesis.

Dynamic description	Reference-subspace hypothesis
Token state changes by context and layer	State is projected relative to $\mathcal{R}_t$
Layer activation carries context	$\mathcal{R}_t^\perp$ contains sense variation
Polysemy by vector deformation	Polysemy by context-conditioned subspaces
Resource is depth and width	Resource is quasi-dimensionality
Limit is activation bottleneck	Limit is interference/accessibility

Second, the spanning matrix $U_{t,C}$ needs to be well-conditioned. If ρ in Eq. (21) is close to one, its dual amplifies readout noise.

Third, the reference-subspace pattern may simply be descriptively wrong. Trained models may represent polysemy without any preferred token-associated direction or subspace. In that case the capacity/accessibility account may still hold, but the reference-subspace hypothesis fails.

Fourth, the law $\sigma_{\text{int}} \sim \sqrt{k/d_{\text{eff}}}$ assumes random-like incoherent directions and weakly correlated signs or amplitudes. Learned networks may improve on this by orthogonalizing frequently coactive features, or worsen it through anisotropy and correlated errors. The law is a scaling principle, not a universal equality.

Finally, the model has choices: the candidate reference subspace $\mathcal{R}_t$, the gate, the selector, and the sense-subspace dimension. These choices must be fixed or controlled empirically. Otherwise the framework risks describing whatever geometry is found after the fact.

D. Empirical predictions

The key empirical task is to separate the general capacity claim from the stronger reference-subspace claim.

Claim A: privileged token reference subspace. For a polysemous token t, there exists a privileged token-associated subspace $\mathcal{R}_t$ such that sense-relevant variation is better exposed in $\mathcal{R}_t^\perp$ than in the orthogonal complements of matched control subspaces of the same dimension.

Claim B: low-dimensional sense subspaces. Within $\mathcal{R}_t^\perp$, sense-conditioned states concentrate near low-dimensional subspaces with small cross-sense coherence.

These claims can fail independently. Claim A can hold while sense information remains high-dimensional. Claim B can hold relative to some non-token-associated subspace, in which case Claim A fails.

The following tests separate the possibilities.

1. *Reference-subspace test.* For a polysemous token, choose candidate reference subspaces generated by the normalized static input embedding, mean contextual embedding, learned token-specific directions, or a low-dimensional combination of these. Project layer states as in Eq. (28). Sense labels should be better separated in $\mathcal{R}_t^\perp$ than in orthogonal complements of random, frequency-matched, or layer-matched control subspaces of the same dimension. The statistic may be classification accuracy or mutual information with sense labels, assessed by permutation over sense labels and bootstrap over token instances.

2. *Operational subspace test.* For each sense s, collect projected states $z_\ell(t,x) \in \mathcal{R}_t^\perp$ and estimate a low-rank subspace by PCA or probabilistic factor analysis. With $U_{t,s}$ the first q principal directions for sense s, measure cross-sense coherence $\|U_{t,s}^\mathsf{T} U_{t,s'}\|_{\text{op}}$. True sense labels should yield lower cross-sense coherence than shuffled labels and stronger separation than matched control reference subspaces.

3. *Gated-readout test.* Linear probes trained on $z_\ell(t,x)$ should perform comparably to probes trained on the full state $h_\ell(t \mid x)$, while probes restricted to the reference component $P_t h_\ell(t \mid x)$ should perform near chance on sense classification.

4. *Overlap-distribution test.* If a representation exploits quasi-dimensionality, pairwise inner products among token embeddings or SAE feature directions should concentrate near zero with variance $\sim 1/d_{\text{eff}}$. Deviations quantify anisotropy and effective dimension loss.

5. *Synthetic concurrency degradation.* At a fixed layer, inject controlled superpositions
$$h_k = h_0 + \lambda \sum_{j=1}^{k} x_j w_j, \tag{34}$$
using directions w_j obtained independently, for example from SAE decoder directions or frozen linear probes. As k increases, readout should degrade smoothly on the scale $\sqrt{k/d_{\text{eff}}}$ for random-like directions, with weaker degradation for feature sets that training has orthogonalized.

6. *Coactivation-dependent geometry.* Frequently coactive features should show reduced mutual coherence relative to rarely coactive features. This tests the broader capacity/accessibility framework through Eq. (33), independently of the reference-subspace hypothesis.

E. Scope of the quantum analogy

The analogy to quantum contextuality is structural and limited. Both settings involve vectors whose operational role depends on a context or basis. But quantum contextuality has the force of no-go theorems such as Kochen–Specker [32, 33]; no analogous theorem is claimed here for semantic embeddings. The borrowed idea is only the pattern of sharing: context-specific bases may include vectors spanning a common subspace, and readout depends on the selected context. A basis is a set of vectors; it is not itself a vector or a subspace. In quantum terminology a complete basis is sometimes called a frame; that usage is distinct from the token reference subspace $\mathcal{R}_t$ defined here.

Three levels of analysis should not be conflated.

1. *Classical high-dimensional representation.* A present-day Transformer carries real-valued hidden states through attention, nonlinear feed-forward maps, residual connections, and normalization. Concentration, quasi-orthogonality, feature superposition, dynamic embeddings, and finite readout are all available at this level. None requires quantum probability or quantum hardware.

2. *Quantum-like semantic or contextual formalism.* States, contexts, bases, interference terms, noncommuting questions, or measurement-like updates may be used as an abstract model of meaning even when the implementation remains classical. The relevant issue is not the vocabulary but whether this extra structure improves compression, explanation, or prediction relative to an ordinary contextual embedding model.

3. *Genuine quantum machine learning or quantum attention.* Here semantic states would be encoded in qubits or other physical quantum degrees of freedom, intermediate evolution would be unitary, phase would affect interference, and measurement would follow quantum probability. A standard Transformer does none of these things.

Most of the present chapter belongs to the first level. The capacity/accessibility distinction is a claim about classical geometry, and the reference-subspace hypothesis is an empirically testable proposal about the internal organization of a classical model. Its use of a common subspace across alternative contexts creates a structural correspondence with the second level, but the correspondence would become a specifically quantum-like model only after additional ingredients—for example, noncommuting context updates or phase-sensitive interference—were formally specified and shown to make discriminative predictions. The tests in Sec. V D therefore target the reference-subspace hypothesis itself; they do not constitute evidence for physical quantum computation.

The dimension of $\mathcal{R}_t$ is also explicit. In Hilbert-space quantum logic, different contexts can share one observable, as in the familiar tripod-like case, or several observables; a four-dimensional example has two contexts with two common observables [13]. The semantic analogue is $\dim(\mathcal{R}_t) = p$: $p = 1$ is the simplest case, while $p > 1$ permits several token-associated reference directions. In every case, sense-specific information is hypothesized to occupy subspaces of $\mathcal{R}_t^{\perp}$.

VI. CONCLUSION

High-dimensional representations can host many semantic possibilities, but finite readout limits how many can be made simultaneously accessible. The main quantitative content is the same pair of scalings introduced in Eqs. (1) and (2):

$$N \lesssim \exp(c\, d_{\text{eff}}\, \varepsilon^2), \qquad \sigma_{\text{int}} \sim \sqrt{k/d_{\text{eff}}}. \tag{35}$$

The first expression is a coexistence bound; the second is a readout noise scale. They are not the same kind of theorem, but together they describe the difference between semantic capacity and semantic accessibility.

Within this framework we proposed the reference-subspace hypothesis: some polysemous tokens may be organized around stable token-associated reference subspaces, with sense information encoded in low-dimensional subspaces of the orthogonal carrier. This hypothesis is not forced by geometry. Its critical test is whether token-associated reference subspaces expose sense structure better than matched controls. A negative result would falsify that organizational hypothesis while leaving the broader superposition and interference picture intact.

Placed in the three-level comparison used throughout the volume, the result is correspondingly bounded. At the first level, a classical Transformer can host a very large repertoire of almost independent semantic directions, contextualize them through attention, and expose only a limited active set at readout. This is the chapter's principal contribution and requires no quantum premise. At the second level, a quantum-like semantic model may represent a state relative to alternative contexts or bases and may add noncommuting updates, interference terms, or a measurement-like selection rule. The reference-subspace construction offers a possible bridge to that language, but in its present form it remains a falsifiable hypothesis about classical LLM representations rather than evidence of nonclassical probability. At the third level, genuine quantum attention would require physical quantum states, unitary evolution, phase-sensitive interference, and terminal quantum measurement. In the terminology of the earlier thought experiment, classical attention mixes values through nonlinear softmax weights, whereas a genuinely quantum mechanism would have to rotate states unitarily before measurement. Present Transformers belong to the former class.

The chapter's relation to the rest of the volume is therefore also a division of labor. Chapter 3 provides the formal vocabulary with which stronger quantum-like claims can be stated; Chapter 7 supplies a classical wave-based comparison in which interference-like behavior arises by a different mechanism; and Chapter 9 asks how representation in LLMs relates to neural systems, embodiment, and meaning. Chapter 13 and Bridge C supply the decisive next step: comparing the reference-subspace model and any stronger quantum-like alternative with classical baselines under conditions where their predictions diverge. Until such tests succeed, the quantum analogy is best treated as a disciplined source of hypotheses, while semantic concurrency and finite contextual readout remain the substantive geometric claims.

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